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Optimizing the LiGen Drug Discovery Pipeline For Intel Max GPUs

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EuroHPC
Joint Undertaking

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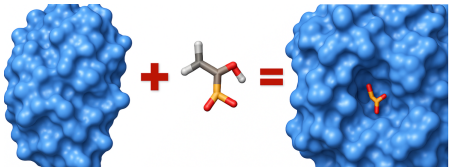
Outline

1. **Motivation & Background**
   Virtual screening, LiGen pipeline, vendor lock-in, GPU hardware diversity
2. **Proposed Approach**
   Performance-portable heuristic (SYCL-heur) + Intel-specific optimizations (SYCL-intel)
3. **Experimental Evaluation**
   Throughput results on Intel Max 1100 & cross-platform roofline analysis
4. **Conclusion**

Motivation & Background

Virtual Screening

- **Ligand**: small molecule (<100 atoms) – drug candidate
- **Protein**: drug target with binding pockets
- **Docking**: fit ligand into protein pocket
  - Protein = rigid body, ligand = flexible
  - Multiple poses per ligand
- **Virtual screening**: evaluate **millions** of candidates computationally **LiGen** [1] – drug discovery platform:
  - Owned by Dompé Farmaceutici
  - Co-developed by PoliMi & CINECA
  - Used for Zika, SARS-CoV-2 drug discovery



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[1] Andrea R. Beccari et al. "LiGen: A High Performance Workflow for Chemistry Driven de Novo Design". In: *Journal of Chemical Information and Modeling* 53.6 (2013), pp. 1518–1527

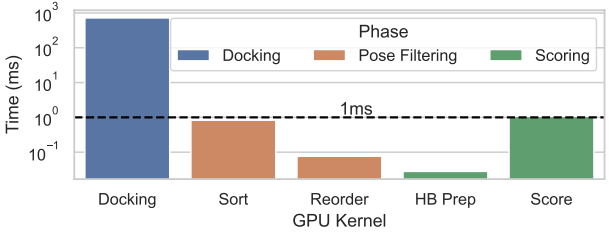
The LiGen Pipeline

- **CUDA**
  - LiGen-Latency
  - **LiGen-Batch**

LiGen-Batch [2], this work's focus:

- Each sub-group processes **one ligand**
- Ligand's atoms are assigned to the subgroup threads
- Multiple ligands per kernel → high throughput

Docking kernel dominates:



Docking: >**99%** of GPU time

⇒ our optimization focus

[2] Emanuele Vitali et al. "GPU-optimized approaches to molecular docking-based virtual screening in drug discovery: A comparative analysis". In: *Journal of Parallel and Distributed Computing* 186 (2024), p. 104819. ISSN: 0743-7315. DOI: https://doi.org/10.1016/j.jpdc.2023.104819

The Portability Challenge

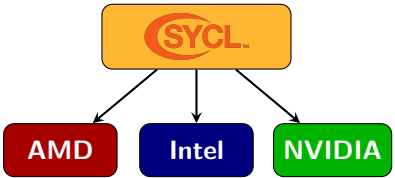
Problem:

- LiGen was **CUDA-only**
- Vendor-locked code cannot target the growing diversity of HPC accelerators

Solution:

- **SYCL** port [3]:
  - LiGen-Latency
  - LiGen-Batch (focus of this work)
- Single-source C++ for all GPU vendors

Portability is solved — but is a single implementation enough for good performance on different architectures?



[3] Luigi Crisci et al. "Enabling performance portability on the LiGen drug discovery pipeline". In: *Future Generation Computer Systems* 158 (2024), pp. 44–59

GPU Hardware Diversity

GPU architectures differ in fundamental ways:

NVIDIA GPUs

- SIMT execution (warps)
- Fixed warp size = 32 threads
- Fixed register allocation
- Tuned for one vendor

Intel Max GPUs

- Explicit SIMD vectorization
- **Configurable sub-group size** (16 or 32)
- **Configurable GRF** (128 or 256 regs)
- Tuning knobs that affect occupancy & spilling

A *portable* SYCL implementation alone is not enough — **architecture-specific tuning** is needed for peak performance.

LiGen SYCL Baseline – Limitations

SYCL solves vendor lock-in, but the baseline **ignores hardware diversity**:

1. No dynamic tuning

- Fixed work-group size (128)
- Fixed ligands per kernel (1120)
- Requires manual tuning per device

2. No Intel-specific opts

- Default sub-group size (32)
- Default GRF mode (auto)
- Misses Intel's configurable knobs

Known issue: LiGen-Batch suffers from **high register pressure**. The dock kernel allocates up to 60% more registers in SYCL than in CUDA, causing significant **register spilling** and limiting occupancy. [3]

Our goal: runtime heuristic for portable tuning + Intel-specific optimizations

[3] Luigi Crisci et al. "Enabling performance portability on the LiGen drug discovery pipeline". In: *Future Generation Computer Systems* 158 (2024), pp. 44–59

Proposed Approach

Contributions

1. **Performance-Portable SYCL Optimizations**

   Runtime heuristic using advanced SYCL/oneAPI features to dynamically adapt kernel workload to the target hardware – *no manual tuning required*

2. **Architecture-Specific Optimizations on Intel GPUs**

   Exploiting sub-group size tuning and GRF configuration to maximize performance on Intel Max GPUs

3. **Comprehensive Experimental Evaluation**

   Comparison of SYCL-baseline, SYCL-heur (portable), SYCL-intel (Intel-optimized), and SYCL-tuned (manual upper bound) with cross-platform roofline analysis against CUDA on NVIDIA V100S

LiGen SYCL Implementations

Version	Work-group	Sub-group	GRF	Lig./Kernel
SYCL-baseline	Fixed (128)	Default (32)	Compiler defined (auto)	Fixed (1120)
SYCL-heur	Heuristic (64–1024)	Default (32)	Compiler defined (auto)	Heuristic defined
SYCL-intel	Heuristic (64–1024)	Heuristic (16, 32)	Heuristic (small, large)	Heuristic defined
SYCL-tuned	Manually tuned (64–1024)	Manually tuned (16, 32)	Manually tuned (small, large)	Manually tuned (1000–4000)

- *SYCL-baseline*: lower bound (manual occupancy calculator)
- *SYCL-heur*: portable, automatic tuning – no hardware expertise needed
- *SYCL-intel*: extends heuristic with Intel-specific features
- *SYCL-tuned*: upper bound (exhaustive manual profiling)

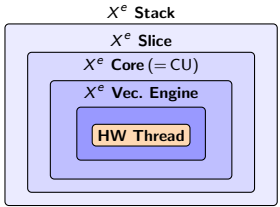
Performance-Portable Heuristic (SYCL-heur)

Goal: Automatically determine optimal # ligands per kernel at runtime.

Core formula [4]: $I = b \times CU \times \frac{wgs}{sgs}$

I = # ligands per kernel
b = max work-groups per CU
CU = compute units (CUDA SMs, Intel X^e Cores)
wgs = work-group size
sgs = sub-group size

Strategy:
Maximize ligand throughput → then minimize register spilling.



[4] Gianmarco Accordi et al. "Out of kernel tuning and optimizations for portable large-scale docking experiments on GPUs".
In: *The Journal of Supercomputing* 80.8 (2024), pp. 11798–11815

Runtime Heuristic Algorithm

Algorithm 1: *SYCL-heur*: Optimizing lig- and throughput

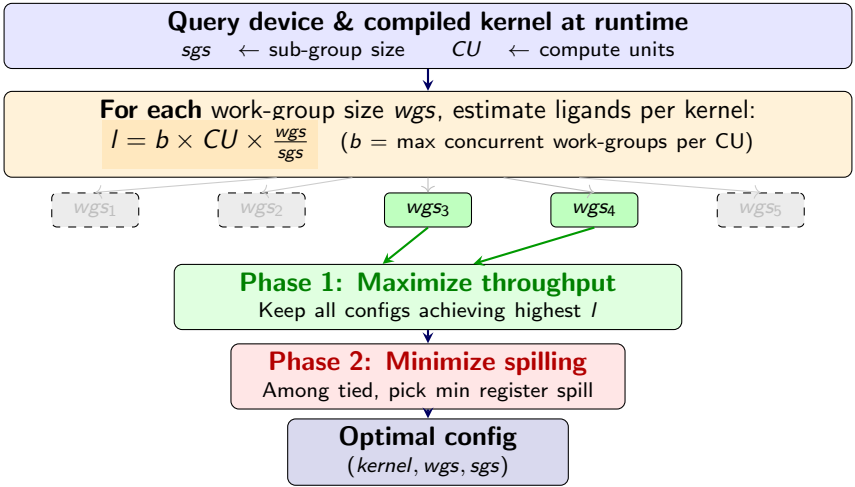
Input: device, slm_per_lig, kernel, wgs_vec
Output: opt_config

1 $sgs \leftarrow kernel.get\_sub\_group\_size()$;
2 $max\_lig\_kern \leftarrow 0$, $configs \leftarrow \emptyset$;
3 $cu \leftarrow device.num\_compute\_units()$;
4 **for** $wgs \in wgs\_vec$ **do**
5 $\quad slm\_wg \leftarrow slm\_per\_lig \cdot (wgs/sgs)$;
6 $\quad lig\_kern \leftarrow max\_wg\_per\_cu(\ldots) \cdot cu \cdot \frac{wgs}{sgs}$;
7 $\quad$ **if** $lig\_kern > max\_lig\_kern$ **then**
8 $\quad\quad max\_lig\_kern \leftarrow lig\_kern$;
9 $\quad\quad configs \leftarrow \{(kernel, wgs, sgs)\}$;
10 $\quad$ **else if** $lig\_kern = max\_lig\_kern$ **then**
11 $\quad\quad configs \leftarrow configs \cup \{(kernel, wgs, sgs)\}$;
12 $opt\_config \leftarrow \arg\min_{c \in configs} c.get\_spilled\_register()$;
13 **return** opt_config;

Key steps:

- **L1** Query default SG size from compiled kernel
- **L3** Query # CUs (portable across vendors)
- **L4–11** For each valid WGS, compute max ligands via the formula; keep best configs
- **L6** `max_wg_per_cu`: estimates concurrent WGs from SLM, registers, HW limits
- **L12** Tiebreaker: minimize **register spilling**

How the Heuristic Works

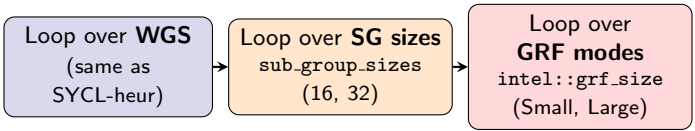


From SYCL-heur to SYCL-intel

SYCL-heur is portable but uses **fixed defaults** on Intel GPUs:

- Default sub-group size (SG=32 on Intel Max)
- Compiler-chosen GRF mode (auto) → not always optimal for register-heavy kernels

SYCL-intel extends the heuristic loop with **two additional dimensions**:

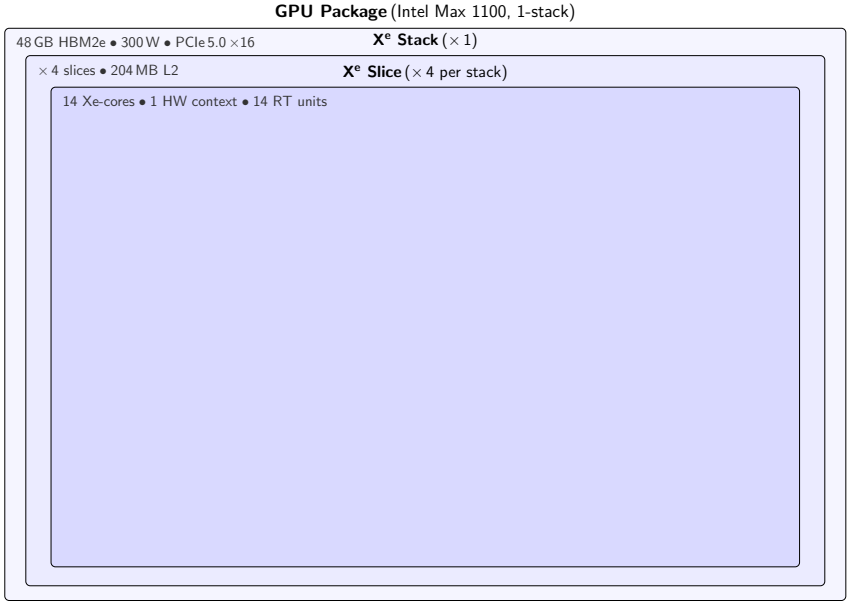


- Kernel **recompiled** for each SG size via `intel::reqd_sub_group_size`
- For each **(WGS, SG, GRF)** combination: evaluate throughput & spilling
- Systematically explores the full configuration space **without manual tuning**

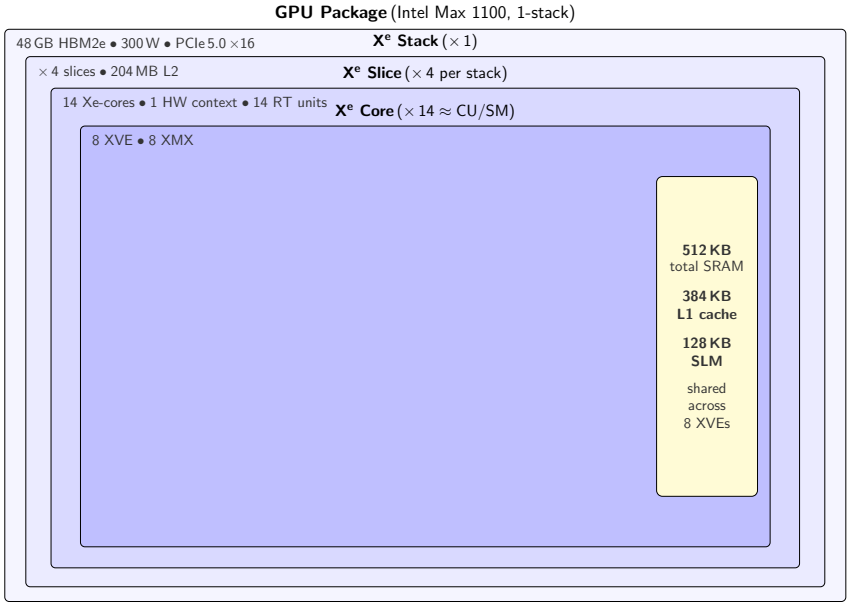
Sub-group Size $\times$ GRF Interplay

Sub-group Size $\times$ GRF Interplay

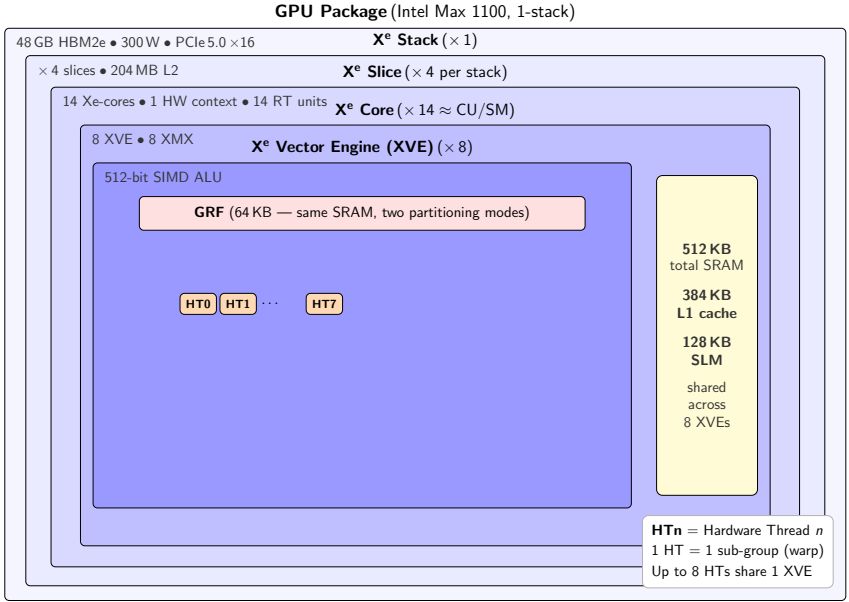
Sub-group Size × GRF Interplay



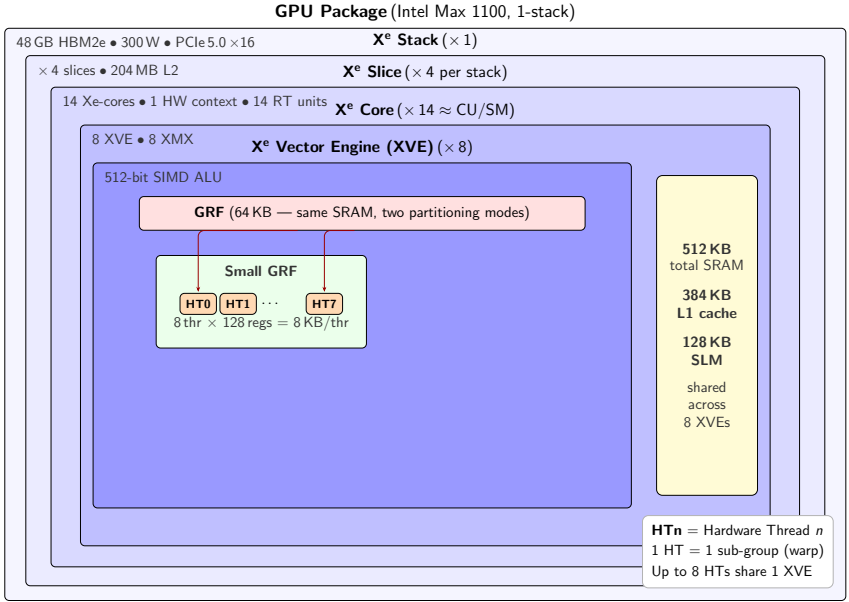
Sub-group Size × GRF Interplay



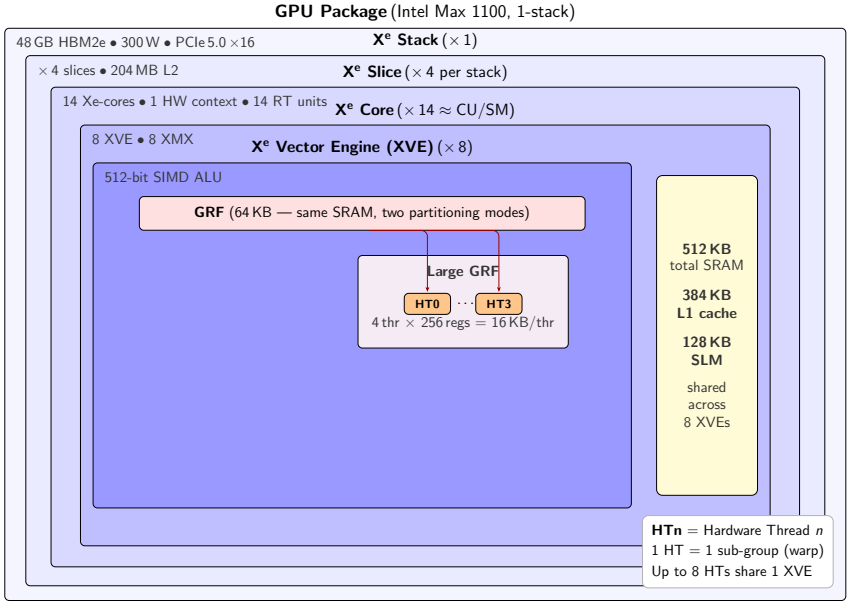
Sub-group Size × GRF Interplay



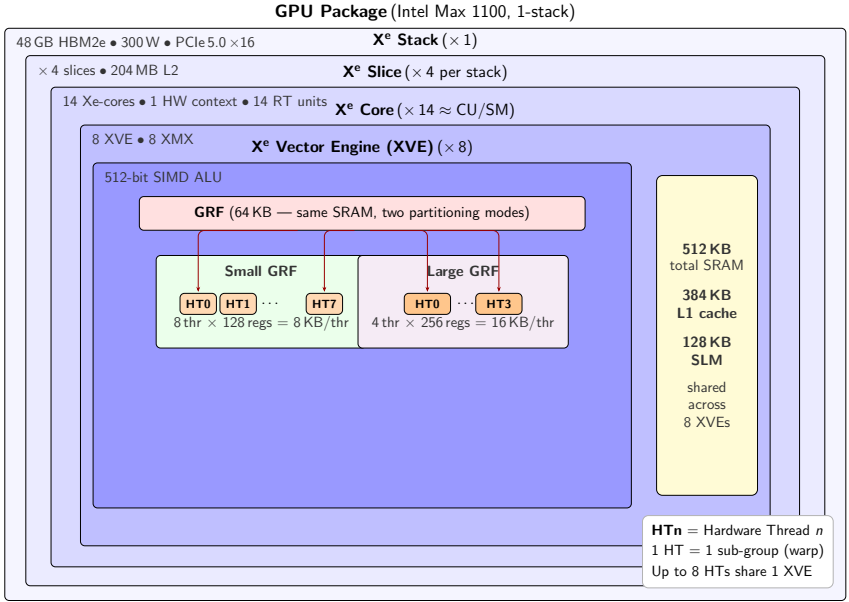
Sub-group Size × GRF Interplay



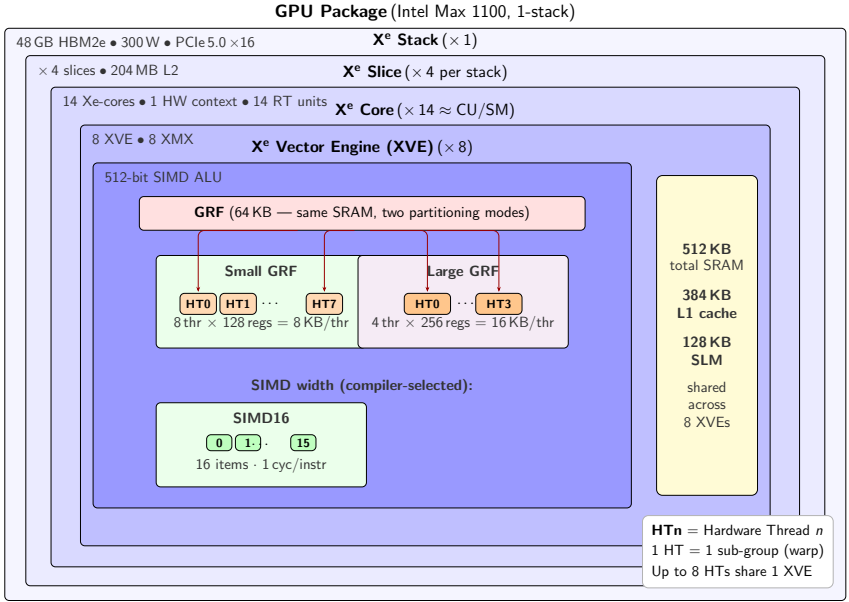
Sub-group Size × GRF Interplay



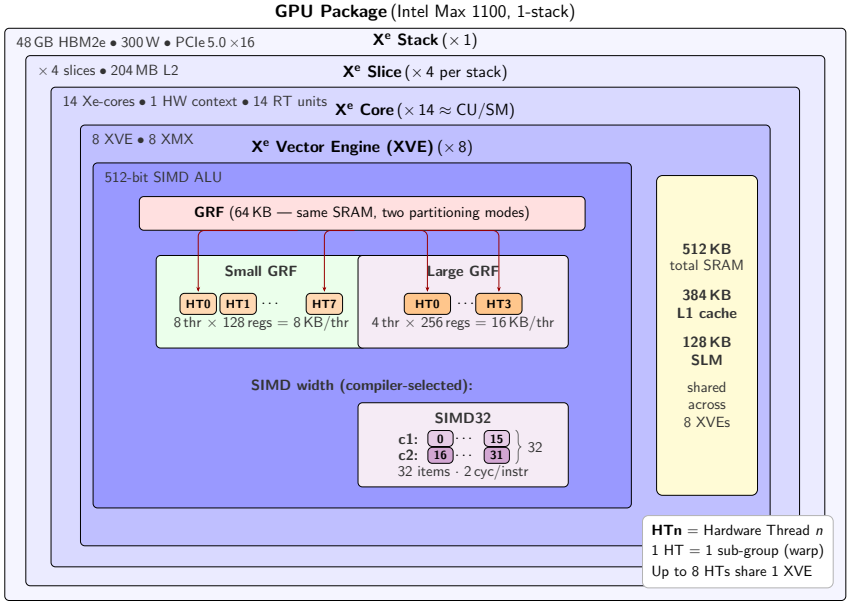
Sub-group Size × GRF Interplay



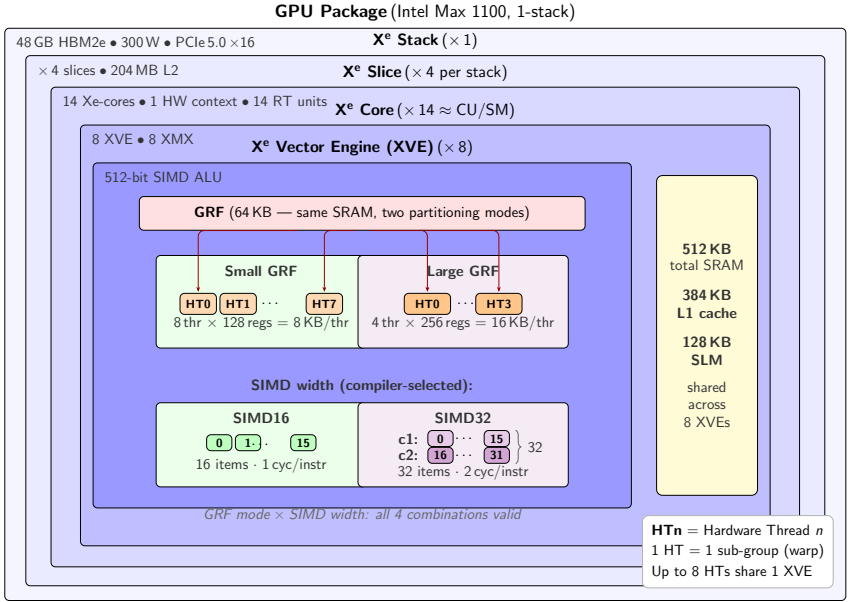
Sub-group Size × GRF Interplay



Sub-group Size × GRF Interplay



Sub-group Size × GRF Interplay



Sub-group Size × GRF Mode on X^e Vector Engine

1. GRF mode — partitions **1024 regs** per XVE among hardware threads:

- **GRF-Small**: 128 regs/HT ⇒ **8 HTs** / XVE *higher occupancy, more latency hiding*
- **GRF-Large**: 256 regs/HT ⇒ **4 HTs** / XVE *halved occupancy, less spilling*

2. SIMD width — sets sub-group size; does **not** change HT count *(each reg = 512 bit)*:

- **SIMD-16**: 16 work-items/HT, each FP32 var = $16 \times 32 = 512$ bits (**1 reg**) *1 cyc/instr*
- **SIMD-32**: 32 work-items/HT, each FP32 var = $32 \times 32 = 1024$ bits (**2 regs**) *2 cyc/instr*

  → **Halves effective variable capacity** with same register budget
  → Doubles work-items in flight

Strategy: Explore all SG × GRF combinations

Experimental Evaluation

Experimental Setup

Hardware

- **SYCL**: Intel Max 1100 GPU (CINECA)
  - Compiler: `icpx` (oneAPI 2025.1.0)
  - Profiler: Intel UniTrace
- **CUDA**: NVIDIA V100S GPU
  - Compiler: `nvcc` (CUDA 12.1)
  - Profiler: NVIDIA Nsight Compute

Datasets

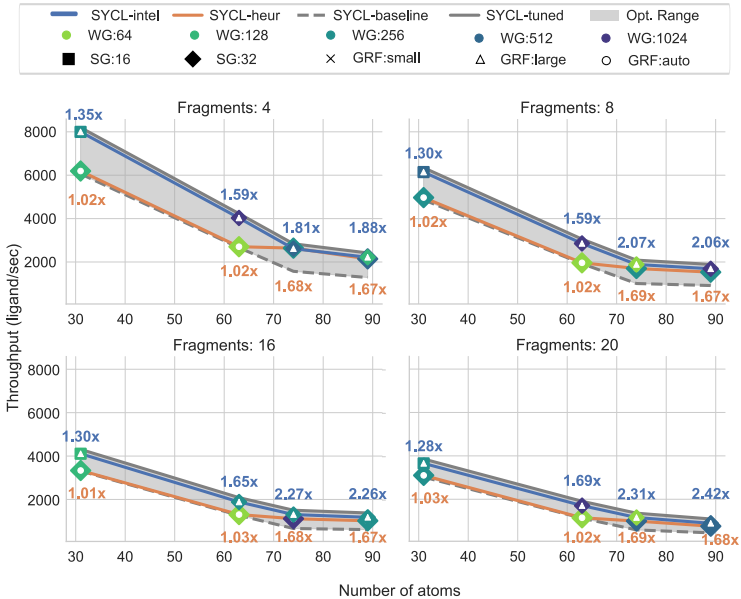
- 100,000 ligands per configuration
- Atom counts: 31, 63, 74, 89
- Fragment sizes: 4, 8, 16, 20
- 3 repetitions per experiment
- Median CoV[1]: 0.15% (stable)

16 configurations (4 atoms $\times$ 4 fragments)

Focus: Docking kernel (>99% of device time)

[1] **CoV:** Coefficient of Variation ($100 \times \frac{\sigma}{\mu}$)

Performance Results



- Ligands/s for different atom-fragment configurations
- Speedups are w.r.t. *SYCL-baseline*
- Gray region: improvement from Intel-specific optimizations

Key Performance Insights

SYCL-heur (portable):

- Geometric mean: **1.31**× vs baseline
- Selects better configs than manual occupancy calculator

SYCL-intel (Intel-optimized):

- Geometric mean: **1.76**× vs baseline
- Best case: **2.42**× (89 atoms, 20 fragments)
- Matches SYCL-tuned upper bound *without profiling*

Why it works:

- Trades occupancy for higher register availability
- Alleviates register spilling (primary bottleneck)
- Improves cache-hit ratio
- Best configs: SG=32 + GRF-Large

Example – 89 atoms, 20 fragments:

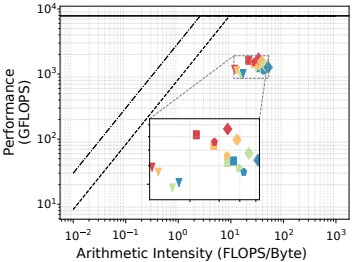
Baseline:	WG=256, 1120 lig/kern
SYCL-heur:	WG=512, 1792 → 1.68×
SYCL-intel:	+GRF-large → **2.42**×

Roofline Analysis

CUDA

NVIDIA V100S

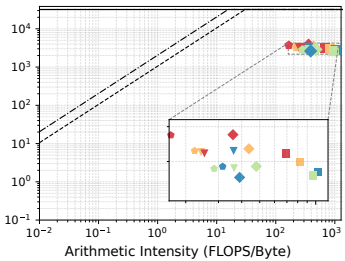
RE[1]: **16.4%**



SYCL-heur

Intel Max 1100

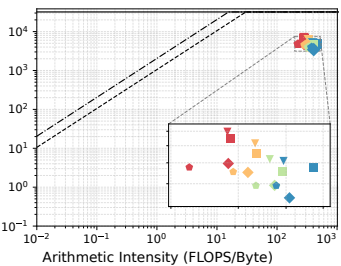
RE: **11.7%**



SYCL-intel

Intel Max 1100

RE: **21.8%**



GPU Memory Roof — L3 Roof — Compute Roof

Fragments: 4 — Fragments: 8 — Fragments: 16 — Fragments: 20

Atoms: 31 — Atoms: 63 — Atoms: 74 — Atoms: 89

[1] RE: Roofline Efficiency (geometric mean)

Conclusion

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Summary:

- Portable heuristic (*SYCL-heur*) dynamically adapts kernel workload at runtime
  - Considers occupancy, register spilling, work-group size
  - Consistently outperforms baseline **without manual tuning**
- Intel-specific optimizations (*SYCL-intel*) deliver additional gains
  - Sub-group size tuning + GRF utilization $\rightarrow$ up to **2.42**$\times$
  - Closely matches manually-tuned upper bound
- Roofline efficiency: SYCL-intel (**21.8%**) $>$ CUDA (**16.4%**)

Key takeaway: Balance between **performance portability** and **architecture-specific optimizations** enables high performance without manual tuning.

Thank you! Questions?